

Department of Physics, IIT-Kanpur

Instructor: Tarun Kanti Ghosh

Quantum Mechanics-II (PHY626)

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Homework-III

(Time-dependent perturbation theory, Rabi oscillation)

1. A one-dimensional harmonic oscillator (with oscillator frequency ω) is subjected to the following force

$$\mathbf{F}(t) = F_0 \frac{\tau/\omega}{t^2 + \tau^2} \hat{\mathbf{x}}, \quad -\infty < t < +\infty.$$

Here τ is a time constant. The time-dependent perturbation is

$$V(t) = \frac{F_0 \tau}{\omega} \frac{x}{t^2 + \tau^2}, \quad -\infty < t < +\infty.$$

The oscillator is prepared in the ground state at $t = -\infty$.

- (a) Sketch $F(t)$ vs ωt for $\omega\tau \ll 1$ and $\omega\tau \gg 1$ on the same graph. Interpret the sketch physically.
- (b) Within the first-order time-dependent perturbation theory, calculate the transition probability to other excited states at $t = \infty$.

$$P_{10} = |c_1^{(1)}(t \rightarrow \infty)|^2 = \frac{(F_0 \pi)^2}{2m\hbar\omega^3} e^{-2\omega\tau}.$$

- (c) Analyse the result for the following limiting cases: (i) $\omega\tau \ll 1$ and (ii) $\omega\tau \gg 1$.
- (d) When $\tau \rightarrow 0$, the time-dependent perturbation can be re-written as

$$V(t) = \frac{F_0 \pi}{\omega} \delta(t).$$

Calculate the transition probability for the δ perturbation and compare with the previous result for $\omega\tau \ll 1$ case.

2. Consider a one-dimensional quantum harmonic oscillator with the oscillation frequency ω is perturbed during $t = 0$ to $t = T$ and it is given by

$$V(t) = \lambda m \omega^2 x^2, \quad 0 < t \leq T.$$

Here $\lambda < 1$ and T is some time-scale, not necessarily to be the time-period of the oscillator.

- (a) Obtain the allowed transitions from the initial state $|i\rangle$ to the final state $|f\rangle \neq |i\rangle$.
- (b) Show that the first-order transition probability is

$$P_{fi} = \frac{4|V_{fi}|^2}{\hbar^2 \omega_{fi}^2} \sin^2(\omega_{fi} T/2).$$

3. **An adiabatic time-dependent perturbation:** Consider the following adiabatic time-dependent perturbation potential:

$$\begin{aligned} V(t) &= 0, & t \leq 0 \\ &= V_0(1 - e^{-t/\tau}), & t > 0. \end{aligned} \quad (1)$$

Here τ is a constant which controls how fast/slow is the perturbation. At large time ($t \rightarrow \infty$, or $t \gg \tau$) limit, $V(t) \rightarrow V_0$ becomes constant. In other words, the perturbation is slowly varying when τ is very large as compared to the other system time scale $1/\omega_{fi}$: $\tau > 1/\omega_{fi}$, or $\omega_{fi} > 1/\tau$.

Within the first-order time-dependent perturbation theory, show that the transition amplitude is

$$c_f^{(1)}(t) = \frac{\langle f|V_0|i \rangle}{i\hbar} \left[\frac{e^{i\omega_{fi}t} - 1}{i\omega_{fi}} - \frac{(e^{i\omega_{fi}t}e^{-t/\tau} - 1)}{i\omega_{fi} - 1/\tau} \right]. \quad (2)$$

Slow process: For the slowly varying perturbation, $\omega_{fi} > 1/\tau$ and in the long time limit $t \gg \tau$, show that

$$c_f^{(1)}(t) \simeq \frac{\langle f|V_0|i \rangle e^{i\omega_{fi}t}}{E_i - E_f}. \quad (3)$$

This is similar to the expansion coefficient of the first-order correction to the wave function in time-independent perturbation theory.

The transition probability

$$P_{fi} = |c_f^{(1)}(t)|^2 = \frac{|\langle f|V_0|i \rangle|^2}{(E_f - E_i)^2}.$$

4. **Quantum harmonic oscillator subjected to a time-periodic δ -kicks:**

- (a) A quantum harmonic oscillator (with oscillator frequency ω) is subjected to a time-periodic δ -kicks with the periodicity T as given by

$$\mathbf{F}(t) = \frac{F_0}{\omega} \sum_{k=0}^{\infty} p^k \delta(t - kT) \hat{\mathbf{x}}.$$

Here $p < 1$ and the factor p^k is implying that kick is reduced at each time interval. The kick periodicity T not necessarily to be equal to the time-period of the oscillator. The oscillator is prepared in a state $|n\rangle$ at $t = 0$. Within the first-order time-dependent perturbation theory, calculate the transition probability to other states $|m\rangle \neq |n\rangle$ at $t = \infty$. Analyse the result for the three cases: (i) $\omega T \ll 1$, (ii) on-resonance condition: $\omega T = 2\pi$ and (iii) $\omega T \gg 1$.

Hint: use the standard result

$$\sum_{k=0}^{\infty} p^k e^{ik\theta} = \frac{1 - pe^{-i\theta}}{1 - 2p\cos\theta + p^2}, \quad |p| < 1.$$

- (b) Repeat the same problem for the δ kicks applied over a finite time NT as described by

$$\mathbf{F}(t) = \frac{F_0}{\omega} \sum_{k=0}^{N-1} \delta(t - kT) \hat{\mathbf{x}}.$$

Answer:

$$|c_{n \rightarrow (n \pm 1)}|^2 = \frac{F_0^2(n \pm 1)}{2\hbar m \omega^3} \frac{\sin^2(N\omega T/2)}{\sin^2(\omega T/2)}.$$

At the resonance $T = 2\pi/\omega$:

$$|c_{n \rightarrow (n \pm 1)}|^2 = \frac{F_0^2(n \pm 1)}{2\hbar m \omega^3} N^2.$$

5. A hydrogen atom in its ground state is subjected to a time-varying (but spatially uniform) electric field $\mathbf{E}(t)$. Within the first-order time-dependent perturbation theory, calculate the transition amplitude from the ground state $|100\rangle$ to the final states $|2lm_l\rangle$ (for $n = 2$: $l = 0, 1$ and for $l = 1$: $m_l = -1, 0, 1$) for the following two electric fields:

$$(a) \quad \mathbf{E}(t) = E_0 e^{-t/\tau} \theta(t) \hat{\mathbf{r}}, \quad (b) \quad \mathbf{E}(t) = E_0 \cos(\Omega t) e^{-t/\tau} \theta(t) \hat{\mathbf{r}},$$

where $\hat{\mathbf{r}}$ is the unit vector in the radial direction and $\theta(t)$ represents the unit step function.

6. Consider a one-dimensional quantum harmonic oscillator with the oscillation frequency ω is perturbed by

$$V(t) = \frac{1}{2} m \omega^2 x^2 \cos(\Omega t) e^{-t/\tau} \theta(t)$$

Here Ω and τ are positive constants.

- (a) Calculate the probability of transition from the state $|n\rangle$ and to the state $|m\rangle$.
 (b) Find out the selection rules for which the transitions are allowed.
 (c) Analyze the features that may arise when $\Omega \rightarrow \omega$ and $\tau \rightarrow 0$ as well as $\tau \rightarrow \infty$.
7. **Dynamics of a two-level quantum system: An exact solution.** Consider an electron's spin in presence of the magnetic field $\mathbf{B} = B_z \hat{\mathbf{z}}$. The spin dynamics can be described by the Hamiltonian $H_0 = -(g\mu_B B_z/2)\sigma_z$, we have assumed $q = +e$ for the convenience. This is an example of a two-level quantum system with the energy levels $E_0 = -(g\mu_B B_z/2)$ and $E_1 = +(g\mu_B B_z/2)$ ($E_1 > E_0$). The corresponding eigenstates are $|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, respectively. The Hamiltonian can also be expressed as

$$H_0 = \begin{pmatrix} E_0 & 0 \\ 0 & E_1 \end{pmatrix}. \quad (4)$$

- (a) Applying a rotating magnetic field:

$$\mathbf{B}(t) = B_0 [\cos(\omega t + \phi) \hat{\mathbf{x}} - \sin(\omega t + \phi) \hat{\mathbf{y}}],$$

where $|\mathbf{B}(t)| = B_0 \ll B_z$ and ϕ is a constant phase angle. It interacts with the electron's spin. Show that the time-dependent perturbation takes the following form:

$$V(t) = \begin{pmatrix} 0 & \delta e^{i\omega t} \\ \delta^* e^{-i\omega t} & 0 \end{pmatrix}.$$

Here $\delta = |\delta|e^{i(\phi+\pi)} = (g\mu_B B_0/2)e^{i(\phi+\pi)}$ is a time-independent complex number with the constant phase $\phi + \pi$. This perturbation induces a transition between the two states.

- (b) Express the total Hamiltonian $H(t) = H_0 + V(t)$ in terms of the Pauli matrices and 2×2 identity matrix σ_0 .
- (c) Express the total Hamiltonian $H(t)$ in terms of the four matrix operators: $|0\rangle\langle 0|$, $|0\rangle\langle 1|$, $|1\rangle\langle 0|$, $|1\rangle\langle 1|$.
- (d) Obtain the time-dependent perturbation $V(t)$ in the Interaction representation i.e.

$$V_I(t) = e^{iH_0 t} V(t) e^{-iH_0 t} = \begin{pmatrix} 0 & \delta e^{i(\omega - \omega_{10})t} \\ \delta^* e^{-i(\omega - \omega_{10})t} & 0 \end{pmatrix}.$$

Here $\omega_{10} = \omega_1 - \omega_0$.

- (e) The most general state can be written in the Interaction representation as

$$|\psi(t)\rangle_I = \begin{pmatrix} c_0(t) \\ c_1(t) \end{pmatrix} = c_0(t)|0\rangle + c_1(t)|1\rangle,$$

where the coefficients $c_0(t)$ and $c_1(t)$ satisfy the normalization condition $|c_0(t)|^2 + |c_1(t)|^2 = 1$. The initial conditions are given by $c_0(0) = 1$ and $c_1(0) = 0$.

Show that the time-evolution of the expansion coefficients can be obtained as

$$\begin{aligned} c_0(t) &= e^{i(\omega - \omega_{10})t} \left[\cos(\Omega t) - i \frac{(\omega - \omega_{10})}{2\Omega} \sin(\Omega t) \right], \\ c_1(t) &= -\frac{i\delta^*}{\hbar\Omega} \sin(\Omega t), \end{aligned}$$

with the Rabi frequency Ω is given by

$$\Omega = \sqrt{\frac{|\delta|^2}{\hbar^2} + \frac{(\omega - \omega_{10})^2}{4}}.$$

- (f) Show that the probability of finding the system at level $|1\rangle$ at time t is

$$|c_1(t)|^2 = \frac{|\delta|^2}{|\delta|^2 + \hbar^2(\omega - \omega_{10})^2/4} \sin^2(\Omega t)$$

and the probability of finding the system at level $|0\rangle$ at time t is

$$|c_0(t)|^2 = 1 - |c_1(t)|^2.$$

It clearly shows that there is a transfer of probability as time goes on. The amplitude of the probability has the Lorentzian shape.

Off-resonance case: For $\hbar|\omega - \omega_{10}| \gg |\delta|$,

$$|c_1(t)|^2 \simeq \frac{4|\delta|^2}{\hbar^2(\omega - \omega_{10})^2} \sin^2 \left(\frac{(\omega - \omega_{10})t}{2} \right). \quad (5)$$

The probability $|c_1(t)|^2$ will never reach to one since the amplitude of $|c_1(t)|^2 < 1$ at all times.

On-resonance case: In this case, the probability densities are $|c_0(t)|^2 = \cos^2[|\delta|t/\hbar]$ and

$$|c_1(t)|^2 = \sin^2(|\delta|t/\hbar). \quad (6)$$

Maximum transition probability, i.e. unit probability can be achieved at resonance $\omega = \omega_{10}$ and at times t_n satisfying the condition $\sin(|\delta|t/\hbar) = 1$, or

$$t_n = \frac{\hbar}{|\delta|} (n + 1/2)\pi, \quad n = 0, 1, 2.. \quad (7)$$

$$= \frac{2\pi\hbar}{g\mu_B B_0} (n + 1/2). \quad (8)$$

Note that the required time to flip the spin (i.e. transition from E_0 level to E_1 level) does not depend on the energy difference between the levels. In other words, t_n does not depend on the constant magnetic field B_z which creates the two-level system.

Typical experimental parameters: $B_z \sim 1$ T and $B_0 \sim 10^{-6}$ T. $t_{\min} = \pi\hbar/(g\mu_B B_0) \sim 10^{-4}$ sec.

- (g) Plot the $|c_1(t)|^2$ and $|c_0(t)|^2$ as a function of $\omega_x t$ for the two cases: (i) on-resonance and (ii) off-resonance. Here $\omega_x = g\mu_B B_0/(2\hbar)$.
- (h) Obtain the spinor $|\psi(t)\rangle_S$ from $|\psi(t)\rangle_I$.

Answer:

$$|\psi(t)\rangle_S = e^{-iH_0 t/\hbar} |\psi(t)\rangle_I = \begin{pmatrix} e^{-i\omega_0 t} c_0(t) \\ e^{-i\omega_1 t} c_1(t) \end{pmatrix}.$$

- (i) Calculate expectation value of σ in both the representations i.e. Interaction as well as Schrödinger representations.

Answer: For both the representations:

$$\begin{aligned} \langle \sigma_x \rangle &= -\frac{|\delta|}{\hbar\Omega} \left[\sin(\omega t + \phi) \sin(2\Omega t) - \frac{\omega - \omega_{10}}{\Omega} \cos(\omega t + \phi) \sin^2(\Omega t) \right] \\ \langle \sigma_y \rangle &= \frac{|\delta|}{\hbar\Omega} \left[\cos(\omega t + \phi) \sin(2\Omega t) + \frac{\omega - \omega_{10}}{\Omega} \sin(\omega t + \phi) \sin^2(\Omega t) \right] \\ \langle \sigma_z \rangle &= \cos^2(\Omega t) + \left(\frac{(\omega - \omega_{10})^2}{4\Omega^2} - \frac{|\delta|^2}{(\hbar\Omega)^2} \right) \sin^2(\Omega t). \end{aligned}$$

- (j) Plot the expectation values of the Pauli matrices ($\langle \sigma_j \rangle$ with $j = x, y, z$) as a function of $\omega_x t$ for the two cases: (i) on-resonance and (ii) off-resonance. Here $\omega_x = g\mu_B B_0/(2\hbar)$.
- (k) **Perturbative results:** Calculate the transition probability from the state $|0\rangle$ to the state $|1\rangle$ using the first-order perturbation theory and compare with the exact results.

Answer:

$$|c_1^{(\text{per})}(t)|^2 = \frac{4|\delta|^2}{\hbar^2(\omega_{10} - \omega)^2} \sin^2 \left[\frac{(\omega_{10} - \omega)t}{2} \right]. \quad (9)$$

This perturbative result matches with Eq. (5) obtained for the off-resonant case.

Near resonant case, $\omega_{10} \simeq \omega$, we have

$$|c_1^{(\text{per})}(t)|^2 \simeq \left(\frac{|\delta|t}{\hbar} \right)^2. \quad (10)$$

This perturbative result matches with the exact result [see Eq. (6)] with $t \ll \hbar/|\delta|$,

- (1) **Rabi oscillation frequency for the realistic magnetic field: Rotating wave approximation.** For $B_z = 1$ T, $\omega_{10} \sim 10^{11}$ radians per sec. In order to reach nearly-resonant and resonant case, the in-plane magnetic field must rotate with the same frequency $\omega_{10} \sim 10^{11}$, which is not possible.

Consider an oscillating magnetic field $\mathbf{B} = B_x \cos(\omega t) \hat{\mathbf{x}}$ with the frequency ω . This oscillating magnetic field may be viewed as a superposition of a magnetic field rotating in the xy plane clockwise with frequency ω and another magnetic field rotating counter-clockwise with the same frequency, with the phase arranged so that the net effect is in the x -direction. This is similar to get a linear polarization of EM radiation out of the sum of two circular polarizations. Only one of the components will travel in the same direction as the precessing spin. The other component will move in a direction opposite to the spin precession and its effect on the electron spin averages out to zero.

Repeat the same problem for $\mathbf{B} = B_x \cos(\omega t) \hat{\mathbf{x}}$ and apply the rotating wave approximation (valid near the resonance). You will get the same results as before.

8. **Dynamics of a three-level quantum system:** Consider a quantum system with just three energy levels E_1 , E_2 and E_3 with corresponding states $|1\rangle$, $|2\rangle$ and $|3\rangle$, respectively. Assuming $E_3 > E_2 > E_1$. The Hamiltonian can be expressed as

$$H_0 = \begin{pmatrix} E_1 & 0 & 0 \\ 0 & E_2 & 0 \\ 0 & 0 & E_3 \end{pmatrix} \quad (11)$$

Applying a time-dependent perturbation of the following form:

$$V(t) = \begin{pmatrix} 0 & \delta^* e^{-i\omega t} & 0 \\ \delta e^{i\omega t} & 0 & \delta^* e^{-i\omega t} \\ 0 & \delta e^{i\omega t} & 0 \end{pmatrix}.$$

Here $\delta = |\delta|e^{i\phi}$ is a time-independent complex number with the constant phase ϕ . This perturbation induces a transition among the three states.

- First obtain the eigenvectors $|n\rangle$ with $n = 1, 2, 3$, correspond to the energy eigenvalues E_n for H_0 .
- Express the total Hamiltonian $H(t) = H_0 + V(t)$ in terms of the spin-1 matrices and 3×3 identity matrix S_0 .
- Express the total Hamiltonian $H(t)$ in terms of the nine matrix operators: $|1\rangle\langle 1|$, $|1\rangle\langle 2|$, $|1\rangle\langle 3|$, $|2\rangle\langle 1|$, $|2\rangle\langle 2|$, $|2\rangle\langle 3|$, $|3\rangle\langle 1|$, $|3\rangle\langle 2|$, $|3\rangle\langle 3|$.
- Obtain the time-dependent perturbation $V(t)$ in the Interaction representation i.e.

$$V_I(t) = e^{iH_0 t} V(t) e^{-iH_0 t}.$$

- (e) The most general state can be written in the Interaction representation as

$$|\psi(t)\rangle_I = c_1(t)|1\rangle + c_2(t)|2\rangle + c_3(t)|3\rangle,$$

where the coefficients $c_1(t)$, $c_2(t)$ and $c_3(t)$ satisfy the normalization condition $|c_1(t)|^2 + |c_2(t)|^2 + |c_3(t)|^2 = 1$.

- (a) Obtain coupled equations of motion of the expansion coefficients.
 - (b) The initial conditions are given by $c_1(0) = 1$ and $c_2(0) = c_3(0) = 0$. Check if you can solve the time-evolution of the expansion coefficients exactly or not.
 - (c) Within the first-order time-dependent perturbation theory, calculate the transition probabilities $P_{21}(t)$, $P_{31}(t)$ and $P_{32}(t)$. [Convention: $P_{f \leftarrow i} = P_{fi}$]
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